Mamba State Space Model with Spectral Feature Conditioning for State of Health Prediction and Anomaly Detection in Lithium-Ion Batteries

Phuc Duy Nguyen, Phuoc Thang Bui, Hong Thai Mai, Minh Tri Tran, and Nhat Minh Nguyen

Faculty of Software Engineering, FPT University Ho Chi Minh City, Ho Chi Minh City, Vietnam

duynpse184821@fpt.edu.vn

thangbpse180445@fpt.edu.vn thaimhse183923@fpt.edu.vn

tritmse183109@fpt.edu.vn minhnnse170310@fpt.edu.vn

Abstract. The fast development of solar energy systems and electric vehicle applications requires robust monitoring of the lithium-ion batteries, since the failure to detect capacity degradation can cause huge economic losses and safety concerns. Although the State of Health (SOH) is the most important indicator to determine the condition of the battery, accurate and real-time estimation from raw sensor data is still a big computational and methodological challenge. Conventional recurrent models struggle to learn long-term temporal dependencies in degradation cycles and Transformer architectures are prohibitively expensive for deployment in edge environments. In this paper, we propose MambaSOHPredictor, a lightweight framework based on the Selective State Space Model (Mamba) with physics-informed spectral feature conditioning. The architecture is available in two configurations, a high-accuracy long-sequence model ($L = 4096$) for full discharge cycles or a compact 30-step window variant for real-time edge inference. MambaBlocks processes the multivariate time-series data (voltage, current, temperature) with linear $O(L)$ complexity. Feature-wise Linear Modulation (FiLM) is used with a 57-dimensional cycle-level spectral and statistical feature vector. Uncertainty estimation is done by Monte Carlo Dropout and anomaly detection is done by Isolation Forest in the same feature space at the same time. Combining the outputs gives us a risk matrix to prioritise predictive maintenance. Long-sequence model under a strict cross-battery protocol yields MAE of 1.52% and RMSE of 1.97% on the NASA Ames dataset surpassing industrial thresholds. The compact configuration can complete inference in less than 100 ms on CPU, which demonstrates practical feasibility for real-time edge deployment without GPU acceleration.

Keywords: Lithium-ion batteries · State of Health (SOH) · Mamba State Space Model · Spectral feature conditioning · FiLM · Anomaly detection · Edge computing · Predictive maintenance.

1 Introduction

It is a fast-moving world of sustainable transport and decarbonisation [8]. Lithium-ion batteries (LIBs) are the most favoured energy storage technology for electric vehicle (EV) and grid-scale renewable integration applications. Road transport accounts for about 15.9% of global greenhouse gas emissions, mainly from passenger cars, and should be strongly electrified [4]. Policy initiatives from the EU such as the "Fit for 55" package and the "Electric 2035" vision are supportive of the LIB market as well. The LIBs market demand is expected to grow by almost 7 folds during 2022–2030 [17]. Complex ageing mechanisms, including solid-electrolyte interphase growth, active lithium depletion and electrode structure deterioration, lead to capacity fade and internal resistance rise, threatening the reliability and safety of long-term LIB systems [10,15].

The trend of battery health monitoring has been changed from physics-based methods such as equivalent circuit model and electrochemical simulation to data-driven deep learning [5]. Recurrent architectures like LSTM have achieved promising results in modelling degradation trajectories. However, they suffer from the recurrence bottleneck in modelling long sequences [22,23]. This problem is addressed by transformer-based models with self-attention but are quadratic in complexity $O(L^2)$ limiting their real-time deployment [13]. Notably, Mamba and selective state space models break this trade-off and reach the performance of Transformers with $O(L)$ linear complexity [6]. But then there's a big hole. The majority of Mamba-based batteries are unimodal studies that only utilise macro-telemetry and overlook the spectral features in the frequency domain that are sensitive to the internal degradation fingerprint [1,10,12].

Moreover, the existing methods mainly focus on the individual estimation of SOH and anomaly detection, which is not sufficient for real-time detection of anomalous patterns due to transient events or sensor anomalies [3]. Furthermore, few effective conditioning mechanisms exist to bridge the distribution gap between high-dimensional temporal sequences and low-dimensional spectral features [10]. Building upon prior work, we propose MambaSOHPredictor, a Selective State Space Model conditioned on physics informed spectral features through Feature-wise Linear Modulation (FiLM). The architecture is amenable to edge implementations, and supports efficient and robust SOH prediction and simultaneous Isolation Forest anomaly detection, enabling reliable real-time battery health management.

2 Related Work

2.1 Deep Sequence Modeling: From RNNs to Mamba

In Section 1, we present an overview of the general path from recurrent architectures to Transformers and selective state space models. This evolution has resulted in several Mamba-based implementations (MambaLithium, SambaMixer and iMamba variants) in battery prognostics claiming better accuracy and efficiency for unified SOH, SOC and RUL estimation than recurrent and attention-based baselines [7,12].

2.2 Health Indicators and Multimodal Frequency Conditioning

These can be telemetry values such as raw voltage/current/temperature or more degradation-sensitive values such as incremental capacity and differential voltage analysis [18], electrochemical impedance spectroscopy [1], and internal gas pressure [19] enabling multi-feature approaches instead of only time-domain ones [21]. It is difficult to fuse such multi-domain information with high-dimensional temporal sequences because of mismatched statistical distributions and manifolds. Modal conflicts and suboptimal gradient flow become common consequences of naive concatenation [10]. Architectures like Hybrid Sensing Synergy Architecture (HSSA) use Q-Former and cross-attention modules to align EIS features to the temporal representation space [10], and Feature-wise Linear Modulation (FiLM) provides a lighter approach to adapt sequence representations directly from spectral conditioning without specialised EIS hardware [7].

2.3 Robust Diagnostics and Anomaly Detection

In order to design a robust and safe AI powered Battery Management System [3], it is needed to estimate the point-wise state-of-health (SOH), quantify uncertainty, and detect anomalies using state-of-the-art models. Anomalies can be caused by sensor noise, distributional shifts or new internal faults and result in accelerated degradation. They are detected by anomaly detection [3]. Hybrid methods have been successfully applied to practical solar-grid and EV cases such as statistical methods such as local outlier factor (LOF) and Mahalanobis distance, and advanced frameworks such as Markov-constrained Isolation Forest [2]. The deployment of these powerful diagnostic tools in efficient Mamba based architectures [10,12] marks a milestone in the development of self-aware, responsible and deployable battery health monitoring systems.

2.4 Literature Comparison Matrix

Table 1. Comprehensive Literature Comparison Matrix and Structural Comparison

Study	Method Family	Key Strength	Key Limitation	Relation to This Work
LSTM / CNN-LSTM [3,23]	Sequential / Hybrid CNN-RNN	Effective temporal modeling on short windows; simple implementation	Recurrence bottleneck; struggles with long-range dependencies and cross-battery generalization	Served as primary baseline for comparison; outperformed by proposed Mamba architecture
Transformer / PatchTST [13]	Attention-based Transformer	Strong global context capture via self-attention	Quadratic complexity $O(L^2)$; high memory/latency on long sequences ($L = 4096$); edge deployment challenges	Advanced baseline demonstrating limitations addressed by linear Mamba backbone
Unimodal Mamba (SambaMixer, Mamba-Lithium) [12]	Selective State Space Model (SSM)	Linear complexity $O(L)$; efficient long-sequence modeling	Unimodal (raw telemetry only); limited physics-informed conditioning and interpretability	Core backbone adopted and extended; this work adds spectral conditioning via FiLM
HSSA (Mamba + EIS Multimodal) [10]	Mamba + Q-Former Multimodal Fusion	Multimodal fusion of discharge curves and impedance spectra; strong performance on NASA	Requires EIS data (not always available in field); higher complexity for edge devices	Closely related multimodal inspiration; this work uses spectral features (FFT/statistical) as more lightweight alternative conditioning signal

Source: Summarize of Authors, 2026

3 Methodology

3.1 Dataset and Evaluation Protocol

The present work utilises the lithium-ion battery lifetime dataset from NASA Ames Prognostics Center of Excellence [16]. The dataset consists of 18650-type cells with a nominal capacity of 2.0 Ah cycled until the capacity falls below a safety threshold. Each discharge cycle has multivariate time-series data (voltage, current, surface temperature) and the end-of-cycle measured capacity. State of Health (SOH) is defined as current capacity over nominal capacity (2.0 Ah) in percentage form.

Quality filtering resulted in the retention of 26 of the original 34 cells. Excluded cells were B0036 (excessive noise), B0049–B0052 (incomplete sequences) and B0038–B0040 (reserved). We further removed cycles with missing capacity values or with too short length. This work uses a strict cross-battery split protocol, which differs from previous work that usually splits data by time step within the same cell. Specifically:

- **Training set:** 23 cells,
- **Validation set:** 2 cells (B0046, B0047),
- **Testing set:** 1 cell (B0048), completely isolated from training.

All validation and test cells were operated at 4 °C. To support learning in this temperature regime, the training set included several 4 °C cells (B0041, B0045, B0053–B0056). Preprocessing and splitting used a fixed random seed of

42 for reproducibility. For additional statistical robustness, a leave-one-battery-out (LOBO) protocol was applied on the window-30 variant, with scalers refitted in each fold to avoid leakage. Results are presented in Section 4.3.

3.2 Data Preprocessing and Feature Extraction

Each discharge cycle provides four raw channels (voltage, current, surface temperature, time stamp) plus two derived channels: the Incremental Capacity ($\mathrm{d}Q/\mathrm{d}V$) curve, which captures aging through shifts in peak amplitude and position [20], and discharge progress, the normalized accumulated discharged capacity (range $[0, 1]$), providing an explicit positional reference within the cycle. As the battery ages, degradation effects appear across the entire discharge profile, including shortened duration, shifted temperature, and systematic decline of the IC peak near 3.5 V, showing that aging signatures are distributed throughout the cycle rather than confined to narrow windows and motivating the use of the full discharge profile ($L = 4096$) as model input.

All six channels are normalized to $[0, 1]$ using `MinMaxScaler`, fitted exclusively on the training set to prevent data leakage. Full-cycle sequences are resampled to $L = 4096$ steps, while the compact configuration uses non-overlapping 30-step sliding windows for real-time inference (Section 3.4), employing a dedicated set of six channels (four raw measurements, normalized cycle index, and estimated SOC).

Each cycle is additionally summarized into a 57-dimensional spectral-statistical feature vector (10 frequency-domain and 9 time-domain descriptors per channel; full list in the project repository), computed over the full cycle for optimal FFT resolution, shared across sliding windows within the same cycle, and z-score normalized on the training distribution.

3.3 Architecture of MambaSOHPredictor

Core Mamba Processing in Pure PyTorch. The computational core is based on the Selective State Space Model (Mamba) [6] implemented in pure native PyTorch without any special CUDA kernels, allowing cross-platform operation (including Windows) and direct CPU inference.

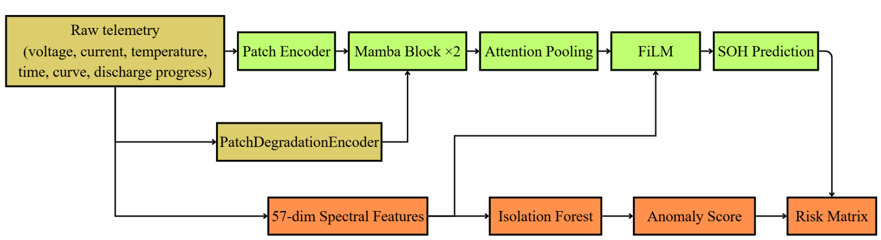

Fig. 1. Overall architecture of the proposed framework

Each MambaBlock consists of an expansion projection ($E = 2$), a depthwise causal convolution (kernel $= 4$), and a selective SSM scan whose state parameters are dynamically adapted from the input [6], which selectively keeps or discards information at each step, suiting battery signals where most of the cycle is stationary and ageing features are concentrated in transitional phases. This leads to linear $O(L)$ complexity for the SSM block compared to the quadratic $O(L^2)$ cost of self-attention at $L = 4096$.

Patch Encoding. The input sequence is divided into patches of 16 steps (stride 8, 50% overlap) and projected to around 511 tokens, similar to PatchTST [11]. Each token is augmented with four local degradation statistics (RMS, peak-to-peak, standard deviation, kurtosis) using the PatchDegradationEncoder, providing dynamic context on top of the global spectral features.

Token Aggregation (Attention Pooling). After two MambaBlocks and LayerNorm, we aggregate the 511 tokens with single-head attention pooling to a single cycle-level representation. This learns a weight distribution for the most informative dynamic phases of the discharge cycle.

Representation Modulation via FiLM. The 57-dimensional spectral vector modulates the pooled representation via FiLM [14]: a two-layer MLP produces scale/shift parameters (γ, β) to transform h as $h \leftarrow (\sigma(\gamma) + 0.5) \odot h + \beta$, which allows the spectral summary to modulate features directly rather than only at the output (Section 5).

Final Regression Head. The aggregated representation is passed through a two-layer head ($64 \to 32 \to 1$, GELU, 0.3 dropout) to output the continuous SOH value.

3.4 Training Setup

The long-sequence configuration is optimized with AdamW (lr 5×10^{-4}, weight decay 3×10^{-4}) against a weighted SmoothL1 loss ($\beta = 0.02$, clipping gradients for residuals above 2% SOH), with near-EOL samples (SOH $<$ 80%) upweighted up to 2×, the weight decaying exponentially with distance from the threshold. Training uses a short warm-up (`CosineAnnealingWarmRestarts`, $T_0 = 3$) followed by a final stage ($T_0 = 80$, $T_{\text{mult}} = 2$) whose periodic restarts escape the plateau seen under fixed decay. The compact window-30 configuration is trained separately with Adam (lr 5×10^{-4}, weight decay 10^{-5}), batch size 32, up to 100 epochs (patience = 15).

3.5 Uncertainty Quantification and Anomaly Detection

Uncertainty Quantification. Predictive uncertainty is estimated via Monte Carlo Dropout: 10 stochastic forward passes (dropout active at inference, batched for latency) yield a mean SOH estimate and standard deviation as the confidence band, with negligible overhead given the compact model's 79,467 parameters.

Anomaly Detection and Risk Management. An Isolation Forest module [9] (100 estimators, contamination = 0.1, seed = 42), fitted on the 57-dimensional spectral feature space, classifies states via `decision_function` into Normal, Warning (≤ -0.1), and Anomaly (≤ -0.3), orthogonal to four SOH-derived levels: Healthy ($\geq 90\%$), Degrading ($\geq 85\%$), Maintenance Required ($\geq 80\%$), End of Life ($< 80\%$), per NASA guidelines.

These two axes are combined with hardware safety thresholds via a rule-based risk matrix that assigns maintenance priority tiers, preserving a decoupled design where deep learning handles long-term trends, Isolation Forest detects sensor disturbances, and physical constraints act as non-overridable safeguards.

4 Experimental Results

4.1 Experimental Setup

Prediction accuracy was assessed using mean absolute error (MAE) and root mean square error (RMSE) in %SOH. Performance evaluation for anomaly detection was done using Precision, Recall and F1-score. The target thresholds were defined by industrial requirements: MAE $< 2\%$ and RMSE $< 3\%$.

All models were trained in a Kaggle GPU environment. Our implementation is in pure PyTorch and does not depend on custom CUDA kernels. Inference latency benchmarks were run on an Intel Core i7-14650HX CPU. For all experiments we employed a fixed random seed of 42 and consistently applied the cross-battery data splitting protocol described in Section 3.1.

4.2 Main Results

Table 2 compares the SOH prediction errors of four models on the held-out test cell B0048 (operated at 4 °C and completely unseen during training). All models were trained using the same 23/2/1 cross-battery split and random seed.

Table 2. SOH Prediction Performance on Held-Out Test Cell B0048 (23/2/1 Cross-Battery Split)

Model	#Params	Input	MAE (%)	RMSE (%)
Naive last-SOH (oracle)†	–	Ground-truth SOH	0.89	1.49
CNN-LSTM (baseline)	61k	telemetry window 30	4.90	6.49
Mamba window-30 (ours)	79k	telemetry window 30	1.98	2.38
Mamba long $L = 4096$ (ours)	99k	telemetry full cycle	1.52	1.97

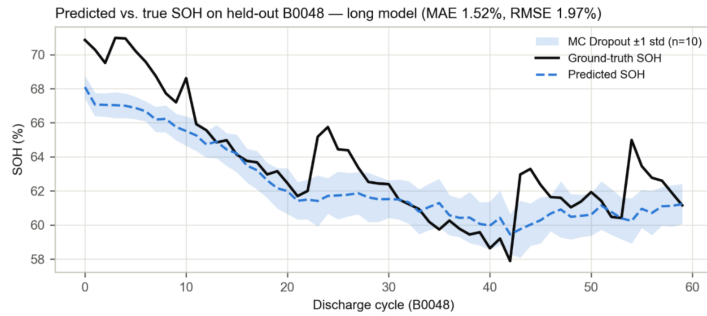

Fig. 2. Predicted vs. ground-truth SOH on B0048 – the sole fully held-out test battery under the cross-battery protocol. Shaded band: ±1 std over 10 MC Dropout samples; per-battery robustness is reported in Table 3 (LOBO)

Among models relying solely on raw telemetry, the realistic deployment scenario, the long-sequence Mamba configuration achieved an MAE of 1.52%, below the 2% industrial target and a 69% relative reduction versus the CNN-LSTM baseline (4.90%), despite the test cell being entirely unseen and operating at an underrepresented 4 °C. The compact window-30 variant likewise reduced relative error by 60% (MAE 1.98%), while CNN-LSTM's limited cross-battery generalization further supports the selective state space choice.

The naive baseline (previous cycle's measured SOH) attained an MAE of 0.89%, outperforming all learned models. We report this honestly, but it is not a viable competing approach since it requires capacity measurements real-world systems cannot obtain every cycle; its value is as an oracle reference for the harder, realistic task addressed above: estimating SOH from telemetry alone. As noted in Section 1, direct comparison with intra-cell split studies is avoided here (see Section 5 for a partial exception).

A data-centric refinement further improved the window-30 model: per-band analysis (Section 4.4) revealed a systematic negative bias in the high-SOH region under 4 °C, where training coverage extended only to 67.2% SOH. Adding one additional 4 °C cell (B0047, covering SOH 0–83.7%), with no architectural or hyperparameter changes, reduced MAE on B0048 from 1.98% to 1.34% (RMSE 2.38% → 1.84%; reported separately due to the modified 24/1/1 split), underscoring that training data coverage can matter as much as architectural choices.

4.3 Robustness Evaluation – Leave-One-Battery-Out

A single held-out test cell cannot establish statistical reliability, so we repeated the experiment via leave-one-battery-out (LOBO) on the window-30 configuration, refitting scalers independently per fold. Of 26 cells, 10 lacked the minimum 60 cycles for reliable evaluation, leaving 16 valid folds.

Table 3. LOBO Robustness (Window-30 Configuration, 16 Folds)

Model	MAE (%)	RMSE (%)
Mean ± std	3.91 ± 3.27	4.81 ± 4.17
Median	2.22	2.55
Fold B0048 (= main battery testing)	1.47	1.87

Source: Authors' processing, 2026

Results reveal considerable variability, so we report the full distribution rather than aggregates alone. Nine of 16 folds achieved MAE $\leq$ 2.3% (best: B0042, 1.40%); the B0048 fold result (Table 3) is consistent with Table 2, confirming its selection was not favorably biased. Larger errors concentrated in cells confined almost entirely to low-SOH regimes: B0045 (14.61%, never exceeding 54% SOH), B0033 (7.55%), B0054–B0056 (4.4–6.3%), forcing extrapolation into SOH regions absent from training data. LOBO thus confirms robustness in typical regimes while revealing a clear failure mode: poor extrapolation to atypical trajectories, discussed further in Section 5.

4.4 Ablation and Error Analysis

(a) Component Ablation. Table 4 quantifies the contribution of individual components by retraining the long-sequence model with each modification while keeping all other conditions unchanged.

Table 4. Component Ablation on Test Cell B0048 (Long Model, 23/2/1 Split)

Variant	MAE (%)	RMSE (%)	Δ MAE
Full model	1.52	1.97	–
Attention pooling $\rightarrow$ last token	1.93	2.38	+0.41
`d_state` 32 $\rightarrow$ 16	1.24	1.61	−0.28
No EOL-weighted loss	1.32	1.69	−0.20

Source: Authors' processing, 2026

Attention pooling contributed most: replacing it with last-token pooling increased MAE by 0.41 percentage points (Table 4), the only degrading change, indicating degradation information is distributed across the sequence rather than concentrated in the final token. Reducing `d_state` (32 $\rightarrow$ 16) and removing the EOL-weighted loss both slightly improved MAE on B0048, but given the single test cell and few near-EOL samples, these gains likely reflect cell-specific variability rather than general superiority, consistent with the LOBO spread (Section 4.3); a full LOBO ablation is left for future work. The final model retains `d_state` = 32 and the EOL-weighted loss, since the larger state dimension aids long-sequence capacity and the weighted loss prioritizes accuracy near the replacement threshold.

(b) Error is not uniform across SOH bands. The 80–90% band shows the largest bias (−4.57%), consistent with reduced near-EOL training density.

4.5 Inference Latency

Latency was measured end-to-end via gRPC (scaler normalization, feature extraction, batched 10-sample MC Dropout inference) across 50 repeated calls on an Intel Core i7-14650HX CPU. The window-30 model achieved a mean of 56.1 ms (P95 = 78.3 ms), satisfying the < 100 ms real-time requirement for P1 alerts; the negligible gap versus direct inference (58.7 ms mean) confirms minimal communication overhead. The long-sequence model trades this real-time capability for higher accuracy, making it more suitable for periodic offline evaluation.

4.6 Anomaly Detection

Because the NASA dataset lacks explicit fault annotations, a rate-based proxy label was defined: cycles exceeding the 90th-percentile training degradation rate (0.4833% SOH/cycle) were considered anomalous, consistent with the Isolation Forest contamination rate (0.1); the model (100 estimators, seed = 42) was trained exclusively on training-set spectral features.

Table 5. Anomaly detection (IsolationForest)

Label	Split	Threshold	Precision	Recall	F1	Positive Rate
rate	val	production (score ≤ -0.1 / -0.3)	0.00	0.00	0.00	35.5%
rate	val	tuned on val (score ≤ 0.213)	0.36	1.00	0.53	35.5%
rate	test	production (score ≤ -0.1 / -0.3)	0.00	0.00	0.00	20.6%
rate	test	tuned on val (score ≤ 0.213)	0.21	1.00	0.34	20.6%

Source: Authors' processing, 2026

The production threshold failed to detect anomalies (F1 = 0); after tuning on the validation set, test F1 improved to 0.34 (100% recall, 0.21 precision), positioning the approach as a high-sensitivity early-warning filter requiring human verification rather than an autonomous fault detector. Since this proxy label reflects capacity degradation rather than genuine faults, future work should incorporate real IoT fault labels and semi-supervised methods (Section 5).

5 Discussion

The 1.52% MAE of the long-sequence model on the strict cross-battery protocol indicates that the selective state space mechanism is able to capture long-range degradation dependencies without the recurrence bottleneck of LSTM-based models or the quadratic cost of Transformers [6,10,23].

Our observation that attention pooling surpasses last-token aggregation aligns with the ablation analysis in Section 4.4, indicating that the signatures of degradation are spread out instead of being localised in a single terminal state during the discharge process, consistent with recent sequence-learning studies [10,11] and confirming our design choice for combining patch encoding with global token aggregation to maintain local and long-range degradation properties.

Table 6. Comparison with Recent Cross-Battery Studies on the NASA Dataset

Work	Architecture	Parameters	Protocol	MAE	Latency	Robustness
SambaMixer-L (2025)	MambaMixer	48.7M	10 train / 3 eval	0.51–1.20%	Not reported	None
TIDSIT (2025)	Transformer	Not reported	2 train / 1 eval	0.58%	Not reported	None
Long model (This work)	Mamba + FiLM	99k	23 train / 2 val / 1 test	1.52%	Not measured (see window-30)	LOBO (16 folds)
Window-30 model (This work)	Mamba + FiLM	79k	Same as above	1.98%	56.1 ms (P95: 78.3 ms)	—

Source: Authors' processing, 2026

The proposed framework does not depend on the injection of electrochemical knowledge using electrochemical impedance spectroscopy (EIS) as was done in previous multimodal Mamba studies [10] but instead extracts FFT-based spectral descriptors directly from discharge telemetry via FiLM conditioning, thus bypassing the need for the specialised measurement hardware required by EIS, while still taking advantage of the higher degradation sensitivity of frequency-domain indicators compared to raw time-domain signals [1,21].

The data-centric improvement shown in Section 4.2, where the addition of a single additional 4 °C battery reduced MAE without any architectural modification, reinforces the notion that operating-condition coverage might be more important than model complexity for cross-battery generalisation, consistent with prior research indicating that performance degradation in battery prognostics is often caused by distribution shift rather than insufficient network capacity [10,21].

This accuracy–latency trade-off compact configuration for real-time edge alerts, long-sequence configuration for periodic cloud diagnostics (Section 4.5) is analogous to recent comparisons between Mamba and Transformer architectures for resource-constrained deployment [12].

As per Section 4.6, the tiny F1 obtained using proxy labels is enough for Isolation Forest as a complementary early-warning module rather than an autonomous fault classifier, a limitation shared by unsupervised anomaly detection in general where proxy labels cannot fully represent real operational failures [2,3].

Three limitations need to be recognised: (i) validation was limited to the NASA dataset (limited cells, single chemistry); (ii) despite the more realistic evaluation, large LOBO performance variation persisted for batteries with rare degradation trajectories; (iii) anomaly detection was based on proxy labels without access to actual fault labels. Therefore, future work should include validation on larger multi-chemistry datasets and real BMS fault logs, as well as exploring semi-supervised or continual learning approaches for better robustness.

6 Conclusion

This paper presents MambaSOHPredictor, a unified Mamba-based architecture evaluated under a cross-battery, cross-temperature protocol substantially more demanding than the timestep-based splits common in the literature. Under this protocol, the long-sequence configuration ($L = 4096$) achieved MAE 1.52% and RMSE 1.97% on the fully unseen, underrepresented-4 °C test cell B0048, surpassing the 2% industrial threshold and reducing relative error by 69% versus a CNN-LSTM baseline. The lightweight window-30 configuration (79k parameters) maintained sub-2% MAE with 56.1 ms mean inference latency (P95 = 78.3 ms) on standard CPU, satisfying real-time constraints for high-priority alerts.

The 16-fold leave-one-battery-out evaluation confirmed B0048 was not a favorably biased choice while revealing a clear failure mode: degraded accuracy for cells confined to rarely observed low-SOH regimes, which we show is mitigated by expanding training data coverage rather than architectural changes. For anomaly detection, we transparently report a modest F1 of 0.34 on statistically meaningful proxy labels, positioning the module as a sensitive early-warning filter requiring human verification rather than an autonomous classifier. Overall, these results demonstrate that industrial-grade accuracy and real-time edge deployability are jointly achievable within a single lightweight, pure-PyTorch framework under a rigorous generalization protocol. Future work will expand training data coverage across underrepresented temperature × SOH combinations, incorporate remaining useful life (RUL) prediction, integrate SOH and anomaly outputs into a maintenance-decision risk matrix, and develop a retrieval-augmented generation (RAG) prescription layer, extensions being pursued within the broader solar battery monitoring system under development by the author group.

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